

DYNAMICS OF BEAM TRUSSES UNDER IMPULSE LOADS

Yves Le Guennec¹, Éric Savin¹, Didier Clouteau²

¹ONERA–The French Aerospace Lab, Châtillon, France; e-mail: yves.le_guennec@onera.fr, eric.savin@onera.fr

²École Centrale Paris, MSSMat Laboratory, Châtenay-Malabry, France; e-mail: didier.clouteau@ecp.fr

Keywords: high frequency, transport equation, discontinuous Galerkin method

Spatial structures are often subjected to impulse loads which induce high-frequency wave propagation. Despite some recent researches, the characterization of the transient response to such loads remains an open problem. These loads can damage the integrity of the structures or equipments attached to them. For example, the solar panels unfolding is set off by a pyrotechnic shock leading to high-frequency wave propagation in the panels. Another example is the pyrotechnic cut for the space launcher/satellite separation. Thus it is necessary to improve the prediction of the high-frequency structural responses with efficient theoretical and numerical models.

In this research, we are interested in high-frequency wave propagation within three-dimensional beam trusses. In the high-frequency range, classical global vibrational methods are not applicable on account of the small wavelength and the high modal density. Therefore two kinds of strategies have been developed by engineers: either global approaches like the Statistical Energy Analysis (SEA) [1] or local approaches like the vibrational conductivity analogy [2]. The main issue of the global approaches is the estimation of some core parameters like the coupling loss factors or the injected powers, whereas the extension of the existing local approaches to built-up systems seems to be difficult. The method adopted here is local and based on the microlocal analysis of classical wave systems. In this framework, it can be proved that the energy density associated with high-frequency elastic waves in a Timoshenko beam satisfy a Liouville-type transport equation [3]:

$$\partial_t w_\alpha + \{\lambda_\alpha, w_\alpha\} = 0, \quad (1)$$

where $w_\alpha(x, k, t)$ is the phase-space energy density of the energy mode α of which eigenfrequency is λ_α , $x \in \mathcal{S}$ is the curvilinear abscissa in the neutral fiber $\mathcal{S} \subset \mathbb{R}$ of the beam, $k \in \mathbb{R}$ is the wave number, and $\{f, g\} = \partial_k f \cdot \partial_x g - \partial_x f \cdot \partial_k g$ is the usual Poisson bracket. This equation highlights the conservation of the vibrational energy in an undamped beam. It is shown in [4] that the energetic modes are separated into two different families: the longitudinal modes that gather the compressional mode and bending modes, and the transverse modes that gather the shear modes and the torsional mode. The longitudinal modes have an eigenfrequency $\lambda_P = |k| \sqrt{E/\rho}$, where E is the Young modulus and ρ is the material density, and the transverse ones have an eigenfrequency $\lambda_T = |k| \sqrt{\kappa\mu/\rho}$, with $\kappa\mu$ the reduced shear modulus. It is remarkable to notice that there is no coupling between modes in the waveguide, an observation that could be somewhat counter-intuitive with an expected diffusive state. In fact all the coupling will occur at the junction.

Indeed, the energy flow is partly reflected and partly transmitted at the interface between substructures. The corresponding reflection/transmission coefficients are derived along the same lines as it is done in [5] for assemblies of two-dimensional beams or plates. The approach is to consider the boundary conditions of the classical wave equation and to derive power flow transmission/reflection coefficients consistent with quadratic observables (wave fluxes). Our results show that the rotational modes (*i.e.* bending and torsion) are totally uncoupled from the translational modes (*i.e.* compression and shear); a mode of one family impinging the junction can not be converted into a mode of the other family. This fact is a remarkable property of the high-frequency dynamics of slender structures.

The objective of this research is to extend the existing model to assemblies of three-dimensional curved beams and develop a reliable model of the high-frequency energy evolution within beam trusses in order to predict, for example, their steady-state behavior at late times or the energy paths. At first, Timoshenko beam theory has been used to describe wave propagation in the truss. The main issue of the proposed derivation is the choice of these kinematics for the beam. Timoshenko hypotheses are indeed not relevant anymore in the high-frequency limit, when the wavelength gets smaller than the cross-section dimensions. This model may be improved by the consideration of Pochhammer-Chree modes in one-dimensional waveguides [7].

As for the numerical resolution of the transport equation (1), classical (continuous) finite element methods can

not be used on account of the discontinuities of the energy density field at the junctions. Thus a Discontinuous Galerkin (DG) finite element method [8] is implemented. The key issue of this method is to introduce numerical fluxes depending on the traces of the physical fluxes at each side of the junction. The spatial discretization is achieved by Legendre polynomials and the direct time integration is performed with a Strong Stability-Preserving Runge-Kutta method [9]. DG methods have the advantages to be weakly dissipative and weakly dispersive depending on their order of interpolation. Thus they are well suited for long times simulation in order to exhibit a potential steady-state behavior. Numerical simulations using the DG method are presented for the example of a three-dimensional beam truss. The numerical implementation and the low-dissipation and low-dispersion properties of our schemes are validated by reversed-time computations. The analysis of these results shows that the steady-state behavior of the truss at late times corresponds to a diffusive behavior as assumed by SEA, for example.

References

- [1] R. H. Lyon, R. G. DeJong, *Theory and Application of Statistical Energy Analysis*. Butterworth-Heineman, Boston MA, 2nd edition, 1995.
- [2] D. J. Nefske, S. H. Sung. Power flow finite element analysis of dynamic systems: basic theory and application to beams. *ASME J. Vib. Acoust. Stress Reliab. Des.*, **111**: 94–100, 1989.
- [3] G. C. Papanicolaou, L. V. Ryzhik, J. B. Keller. Transport equations for elastic and other waves in random media. *Wave Motion*, **24**: 327–370, 1996.
- [4] Y. Le Guennec, É. Savin. A transport model and numerical simulation of the high-frequency dynamics of three-dimensional beam trusses. *J. Acoust. Soc. Am.*, **130**: 3706–3722, 2011.
- [5] É. Savin. A transport model for high-frequency vibrational power flows in coupled heterogeneous structures. *Interaction Multiscale Mech.*, **1**: 53–81, 2007.
- [6] B. Lapeyre, É. Pardoux, R. Sentis. *Introduction to Monte-Carlo Methods for Transport and Diffusion Equations*. Oxford University Press, Oxford, 2003.
- [7] J. D. Achenbach. *Wave Propagation in Elastic Solids*. North-Holland, Amsterdam, 1973.
- [8] J. S. Hesthaven, T. Warburton. *Nodal Discontinuous Galerkin Methods*. Springer, New York NY, 2008.
- [9] S. Gottlieb, C.-W. Shu, E. Tadmor. Strong Stability-Preserving High-Order Time Discretization Methods. *SIAM Review*, **43**: 89–112, 2001.